Unit 2 Quadratic Functions

Expectation

By the end of this course, students will:

- verify, through investigation with and without technology, that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$, $a \geq 0$, $b \geq 0$, and use this relationship to simplify radicals (e.g., $\sqrt{24}$) and radical expressions obtained by adding, subtracting, and multiplying [e.g., $(2 + \sqrt{6})(3 - \sqrt{12})$]
- determine the number of zeros (i.e., x-intercepts) of a quadratic function, using a variety of strategies (e.g., inspecting graphs; factoring; calculating the discriminant)
- determine the maximum or minimum value of a quadratic function whose equation is given in the form $f(x) = ax^2 + bx + c$, using an algebraic method (e.g., completing the square; factoring to determine the zeros and averaging the zeros)
- solve problems involving quadratic functions arising from real-world applications and represented using function notation
- determine, through investigation, the trans- formational relationship among the family of quadratic functions that have the same zeros, and determine the algebraic representation of a quadratic function, given the real roots of the corresponding quadratic equation and a point on the function
- solve problems involving the intersection of a linear function and a quadratic function graphically and algebraically

Homework Log

Lesson	Date	Section	Assigned Work	Extra Practice
1		3.4 Simplifying Radicals Part 1	LP: p. 4 #1 – 7	Text: pg. 167 #1, 2, 4
2		3.4 Simplifying Radicals Part 2	LP: p. 7 – 8 #1 – 12 Text: p. 168 #13, 15 – 17	Text: p. 167 #3, 5 – 12
3		3.2 Determining the Max and Min Values of Quadratic Functions	LP: p. 12 – 13 #1 – 8	Text: p. 153 – 154 #7d, 8 – 10, 11ab
4		3.5 Solving Quadratic Equations	LP: p. 16 #1 – 5	Text: p. 177 #1 – 5
5		3.5 Solving Quadratic Equations – Applications	LP: p. 16 – 17 #6 – 13 Text: p. 178 #17	Text: p. 178 #6 – 16
6		3.6 The Zeros of a Quadratic Function	Text: p. 185 – 186 #4, 7 – 11, 14,15, 17, 18	Text: p. 185 #1 – 3, 5
7		3.7 Families of Quadratic Functions	Text: p. 192 #1 – 4, 6, 8, 9, 10	
8		3.8 Linear-Quadratic Systems	Text: p. 198 – 199 #3, 4, 6, 8 – 12, 14, 16	Text: p. 198 #2, 5, 7, 13, 15
9		Review for Test	LP: p. 24 – 25	Text: p. 202 – 203 #5, 9, 11 – 22

Simplifying Radicals

Recall: $\sqrt{a}$

Review: Solve each of the following.

a) $\sqrt{100 \times 25}$ b) $\sqrt{100} \times \sqrt{25}$ c) $\sqrt{\dfrac{100}{25}}$ d) $\dfrac{\sqrt{100}}{\sqrt{25}}$

Radicals can be simplified using the following properties:

A radical is in simplest form when:

- the radicand has no ____________ ___________ factors other than 1;
- the radicand does not contain a fraction;
- no _____________ appears in the denominator of a fraction.

Example 1: Simplify each of the following.

a) $\sqrt{12}$ b) $\sqrt{54}$ c) $5\sqrt{28}$ d) $\dfrac{\sqrt{40}}{\sqrt{5}}$ e) $\sqrt{\dfrac{2}{9}}$

Multiplying Radicals

Example 2: Simplify each of the following.

a) $\sqrt{13} \times \sqrt{13}$ b) $3\sqrt{2} \times 4\sqrt{7}$ c) $2\sqrt{5} \times 3\sqrt{10}$ d) $2\sqrt{3} \times \sqrt{5}$

Simplifying Radical Expressions

Example 3: Simplify each of the following.

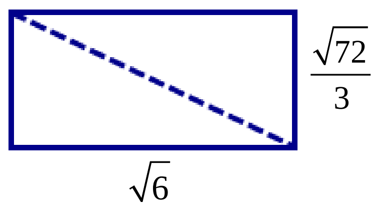
a) $\dfrac{5+\sqrt{75}}{5}$

b) $\dfrac{-12+\sqrt{48}}{4}$

Application Example

Given the rectangle below,

a) Express the exact area of the rectangle in simplest radical form.



b) Express the length of the diagonal in simplest radical form.

Simplifying Radicals Practice

1. Simplify.

a) $\sqrt{12}$ b) $\sqrt{20}$ c) $\sqrt{45}$ d) $\sqrt{50}$ e) $\sqrt{24}$ f) $\sqrt{63}$ g) $\sqrt{200}$ h) $\sqrt{32}$

i) $\sqrt{44}$ j) $\sqrt{60}$ k) $\sqrt{18}$ l) $\sqrt{54}$ m) $\sqrt{128}$ n) $\sqrt{90}$ o) $\sqrt{125}$ p) $\sqrt{80}$

2. Simplify.

a) $\frac{\sqrt{14}}{\sqrt{7}}$ b) $\frac{\sqrt{10}}{\sqrt{2}}$ c) $\frac{\sqrt{60}}{\sqrt{3}}$ d) $\frac{\sqrt{40}}{\sqrt{5}}$ e) $\frac{\sqrt{33}}{\sqrt{3}}$ f) $\frac{\sqrt{7}}{\sqrt{4}}$

g) $\frac{\sqrt{20}}{\sqrt{9}}$ h) $\frac{3\sqrt{8}}{\sqrt{2}}$ i) $\frac{27\sqrt{15}}{3\sqrt{5}}$ j) $\frac{12\sqrt{75}}{4\sqrt{3}}$ k) $\frac{4\sqrt{2}}{\sqrt{8}}$ l) $\frac{2\sqrt{2}}{\sqrt{18}}$

3. Simplify.

a) $\sqrt{2} \times \sqrt{10}$ b) $\sqrt{3} \times \sqrt{6}$ c) $\sqrt{15} \times \sqrt{5}$ d) $\sqrt{7} \times \sqrt{11}$

e) $4\sqrt{3} \times \sqrt{7}$ f) $3\sqrt{6} \times 3\sqrt{6}$ g) $2\sqrt{2} \times 3\sqrt{6}$ h) $2\sqrt{5} \times 3\sqrt{10}$

i) $3\sqrt{3} \times 4\sqrt{15}$ j) $4\sqrt{7} \times 2\sqrt{14}$ k) $\sqrt{6} \times \sqrt{3} \times \sqrt{2}$ l) $2\sqrt{7} \times 3\sqrt{1} \times \sqrt{7}$

4. Simplify.

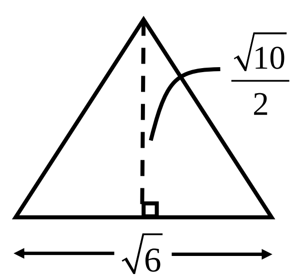
a) $\frac{10+15\sqrt{5}}{5}$ b) $\frac{21-7\sqrt{6}}{7}$ c) $\frac{6+\sqrt{8}}{2}$ d) $\frac{12-\sqrt{27}}{3}$ e) $\frac{-10-\sqrt{50}}{5}$ f) $\frac{-12+\sqrt{48}}{4}$

5. Express each of the following as an integer.

a) $\sqrt{5^2}$ b) $\left(\sqrt{5}\right)^2$ c) $\sqrt{(-5)^2}$ d) $-\sqrt{5^2}$ e) $-\left(\sqrt{5}\right)^2$ f) $-\sqrt{(-5)^2}$

6. Express the exact area of the triangle in simplest radical form.

7. A square has an area of 675 cm^2. Express the side length in simplest radical form.



Answers

1. a) $2\sqrt{3}$ b) $2\sqrt{5}$ c) $3\sqrt{5}$ d) $5\sqrt{2}$ e) $2\sqrt{6}$ f) $3\sqrt{7}$ g) $10\sqrt{2}$ h) $4\sqrt{2}$ i) $2\sqrt{11}$ j) $2\sqrt{15}$ k) $3\sqrt{2}$ l) $3\sqrt{6}$ m) $8\sqrt{2}$ n) $3\sqrt{10}$ o) $5\sqrt{5}$ p) $4\sqrt{5}$ 2. a) $\sqrt{2}$ b) $\sqrt{5}$ c) $2\sqrt{5}$ d) $2\sqrt{2}$ e) $\sqrt{11}$ f) $\frac{\sqrt{7}}{2}$ g) $\frac{2\sqrt{5}}{3}$ h) 6 i) $9\sqrt{3}$ j) 15 k) 2 l) $\frac{2}{3}$ 3. a) $2\sqrt{5}$ b) $3\sqrt{2}$ c) $5\sqrt{3}$ d) $\sqrt{77}$ e) $4\sqrt{21}$ f) 54 g) $12\sqrt{3}$ h) $30\sqrt{2}$ i) $36\sqrt{5}$ j) $56\sqrt{2}$ k) 6 l) 42 4. a) $2+3\sqrt{5}$ b) $3-\sqrt{6}$ c) $3+\sqrt{2}$ d) $4-\sqrt{3}$ e) $-2-\sqrt{2}$ f) $-3+\sqrt{3}$ 5. a) 5 b) 5 c) 5 d) -5 e) -5 f) -5 6. $\frac{\sqrt{15}}{2}$ 7. $15\sqrt{3}$ cm

Simplifying Radicals - Part 2

Radical expressions can be simplified by adding or subtracting _________ ______________.

If a radicand is not in simplest form, always **simplify** it first.

Adding and Subtracting Radicals

Example 1: Simplify each of the following.

a) $3\sqrt{2} - \sqrt{2} + 4\sqrt{2}$

b) $\sqrt{2} + \sqrt{3} - 4\sqrt{3} - 2\sqrt{2} + 5\sqrt{3}$

c) $\sqrt{28} + \sqrt{63}$

d) $\sqrt{20} + \sqrt{40} - \sqrt{45} + \sqrt{90}$

e) $-2\sqrt{20} + 4\sqrt{125} - 3\sqrt{500} + \sqrt{45}$

Multiplying a Radical by a Binomial

Example 2: Simplify each of the following.

a) $\sqrt{5}\left(\sqrt{10} - 1\right)$

b) $3\sqrt{2}\left(2\sqrt{6} + \sqrt{10}\right)$

c) $2\sqrt{2}\left(\sqrt{10} - 3\sqrt{2}\right)$

Binomial Multiplication

d) $\left(\sqrt{3} + \sqrt{2}\right)^2$

e) $\left(6 - \sqrt{2}\right)^2$

f) $\left(3\sqrt{5} + 1\right)\left(3\sqrt{5} - 1\right)$

g) $\left(3\sqrt{7} - 2\sqrt{5}\right)\left(5\sqrt{7} - 4\sqrt{5}\right)$

Recall that a radical is in **simplest form** when:

- the radicand has no perfect square factors apart from 1;
- the radicand does not contain a fraction;
- **no radical appears in the denominator of a fraction**.

Fractions with Radicals in the Denominator

Example 3: Simplify each of the following.

a) $\frac{1}{\sqrt{2}}$

b) $\frac{3\sqrt{2}}{2\sqrt{3}}$

Rationalizing Binomial Denominators

c) $\frac{1}{6+\sqrt{2}}$

d) $\frac{5}{2\sqrt{6}-\sqrt{3}}$

e) $\frac{\sqrt{5}+3}{2\sqrt{5}-4}$

Radicals Practice

1. Simplify.

a) $8\sqrt{7}+2\sqrt{28}$ b) $3\sqrt{50}-2\sqrt{32}$ c) $5\sqrt{27}+4\sqrt{48}$ d) $3\sqrt{8}+\sqrt{18}+3\sqrt{2}$

e) $\sqrt{5}+2\sqrt{45}-3\sqrt{20}$ f) $4\sqrt{3}+3\sqrt{20}-2\sqrt{12}+\sqrt{45}$ g) $3\sqrt{48}-4\sqrt{8}+4\sqrt{27}-2\sqrt{72}$

2. Expand and simplify.

a) $\sqrt{2}\left(\sqrt{10}+4\right)$ b) $\sqrt{3}\left(\sqrt{6}-1\right)$ c) $\sqrt{6}\left(\sqrt{2}+\sqrt{6}\right)$ d) $2\sqrt{2}\left(3\sqrt{6}-\sqrt{3}\right)$ e) $\sqrt{2}\left(\sqrt{3}+4\right)$

f) $3\sqrt{2}\left(2\sqrt{6}+\sqrt{10}\right)$ g) $\left(\sqrt{5}+\sqrt{6}\right)\left(\sqrt{5}+3\sqrt{6}\right)$ h) $\left(2\sqrt{3}-1\right)\left(3\sqrt{3}+2\right)$ i) $\left(4\sqrt{7}-3\sqrt{2}\right)\left(2\sqrt{7}+5\sqrt{2}\right)$

j) $\left(3\sqrt{3}+1\right)^2$ k) $\left(2\sqrt{2}-\sqrt{5}\right)^2$ l) $\left(2+\sqrt{3}\right)\left(2-\sqrt{3}\right)$ m) $\left(\sqrt{6}-\sqrt{2}\right)\left(\sqrt{6}+\sqrt{2}\right)$

n) $\left(2\sqrt{7}+3\sqrt{5}\right)\left(2\sqrt{7}-3\sqrt{5}\right)$

3. Simplify.

a) $\dfrac{1}{\sqrt{3}}$ b) $\dfrac{2}{\sqrt{5}}$ c) $\dfrac{2}{\sqrt{7}}$ d) $\dfrac{\sqrt{1}}{\sqrt{2}}$ e) $\dfrac{5\sqrt{5}}{2\sqrt{3}}$ f) $\dfrac{2\sqrt{2}}{\sqrt{18}}$

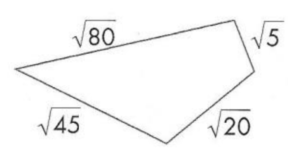
g) $\dfrac{4\sqrt{2}}{\sqrt{8}}$ h) $\dfrac{3\sqrt{5}}{\sqrt{3}}$ i) $\dfrac{4\sqrt{7}}{2\sqrt{14}}$ j) $\dfrac{3\sqrt{6}}{4\sqrt{10}}$ k) $\dfrac{7\sqrt{11}}{2\sqrt{3}}$ l) $\dfrac{2\sqrt{5}}{5\sqrt{2}}$

4. Simplify.

a) $\dfrac{1}{\sqrt{2}+2}$ b) $\dfrac{3}{\sqrt{5}-1}$ c) $\dfrac{\sqrt{2}}{\sqrt{6}-3}$ d) $\dfrac{2}{\sqrt{6}+\sqrt{3}}$ e) $\dfrac{3}{\sqrt{5}-\sqrt{2}}$

f) $\dfrac{\sqrt{3}}{\sqrt{3}+\sqrt{2}}$ g) $\dfrac{2\sqrt{6}}{2\sqrt{6}+1}$ h) $\dfrac{\sqrt{2}-1}{\sqrt{2}+1}$ i) $\dfrac{\sqrt{2}+\sqrt{5}}{\sqrt{6}-\sqrt{10}}$ j) $\dfrac{2\sqrt{7}+\sqrt{5}}{3\sqrt{7}-2\sqrt{5}}$

5. Express the perimeter of the quadrilateral in simplest radical form.



6. Without using a calculator, arrange the following expressions in order from greatest to least.

$$\sqrt{3}\left(\sqrt{3}+1\right),\left(\sqrt{3}+1\right)\left(\sqrt{3}-1\right),\left(1-\sqrt{3}\right)^2,\left(\sqrt{3}+1\right)^2$$

7. a) Without using a calculator, decide which of the following radical expressions does not equal any of the others.

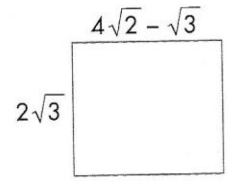
$$\frac{60}{\sqrt{450}}, 6\sqrt{2} - 4\sqrt{2}, \frac{4}{\sqrt{2}}, 6\sqrt{8} - 5\sqrt{8}, \frac{8}{\sqrt{18}} + \frac{4}{\sqrt{18}}$$

   b) How is the radical expression you identified in part a) related to each of the others?

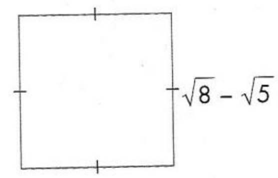
8. Write and simplify an expression for

   a) the area of the rectangle

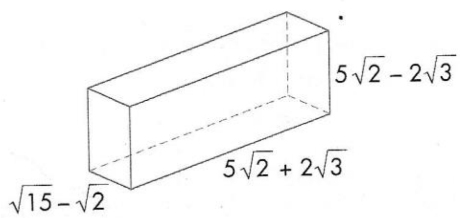
   b) the perimeter of the rectangle



9. Write and simplify an expression for the area of the square.



10. Express the volume of the rectangular prism in the simplest radical form.



11. If a rectangle has an area of 4 square units and a width of $\sqrt{7} - \sqrt{5}$ units, what is its length, in simplest radical form?

Answers

1. a) $12\sqrt{7}$ b) $7\sqrt{2}$ c) $31\sqrt{3}$ d) $12\sqrt{2}$ e) $\sqrt{5}$ f) $9\sqrt{5}$ g) $24\sqrt{3} - 20\sqrt{2}$ 2. a) $2\sqrt{5} + 4\sqrt{2}$ b) $3\sqrt{2} - \sqrt{3}$ c) $2\sqrt{3} + 6$ d) $12\sqrt{3} - 2\sqrt{6}$ e) $\sqrt{6} + 4\sqrt{2}$ f) $12\sqrt{3} + 6\sqrt{5}$ g) $23 + 4\sqrt{30}$ h) $16 + \sqrt{3}$ i) $26 + 14\sqrt{14}$ j) $28 + 6\sqrt{3}$ k) $13 - 4\sqrt{10}$ l) 1 m) 4 n) -17 3. a) $\frac{\sqrt{3}}{3}$ b) $\frac{2\sqrt{5}}{5}$ c) $\frac{2\sqrt{7}}{7}$ d) $\frac{\sqrt{2}}{2}$ e) $\frac{5\sqrt{15}}{6}$ f) $\frac{2}{3}$ g) 2 h) $\sqrt{15}$ i) $\sqrt{2}$ j) $\frac{3\sqrt{15}}{20}$ k) $\frac{7\sqrt{33}}{6}$ l) $\frac{\sqrt{10}}{5}$

4. a) $\frac{2-\sqrt{2}}{2}$ b) $\frac{3+3\sqrt{5}}{4}$ c) $-\frac{3\sqrt{2}+2\sqrt{3}}{3}$ d) $\frac{2\sqrt{6}-2\sqrt{3}}{3}$ e) $\sqrt{5} + \sqrt{2}$ f) $3 - \sqrt{6}$ g) $\frac{24 - 2\sqrt{6}}{23}$ h) $3 - 2\sqrt{2}$

i) $-\frac{5\sqrt{2} + 2\sqrt{3} + 2\sqrt{5} + \sqrt{30}}{4}$ j) $\frac{52 + 7\sqrt{35}}{43}$ 5. $10\sqrt{5}$ 6. $\left(\sqrt{3}+1\right)^2, \sqrt{3}\left(\sqrt{3}+1\right), \left(\sqrt{3}+1\right)\left(\sqrt{3}-1\right), \left(1-\sqrt{3}\right)^2$

7. a) $6\sqrt{8} + \sqrt{8} - 5\sqrt{8}$ b) It is twice as large as the others. 8. a) $8\sqrt{6} - 6$ b) $8\sqrt{2} + 2\sqrt{3}$ 9. $13 - 4\sqrt{10}$ 10. $38\sqrt{15} - 38\sqrt{2}$

11. $2\sqrt{7} + 2\sqrt{5}$

Surds $\sqrt{5}$ $\sqrt{2}$ $\sqrt{17}$

- ☐ Surds are roots of rational numbers that can't be written as rational numbers.
- ☐ They are **irrational** numbers.

Rule 1 $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

Example – Simplify $\sqrt{18}$

$\sqrt{18} = \sqrt{9} \times \sqrt{2}$

$= 3 \times \sqrt{2}$

$= 3\sqrt{2}$

Try it yourself

Simplify $\sqrt{32}$

Rule 2 $m\sqrt{a} + n\sqrt{a} = (m+n)\sqrt{a}$

Simplify $5\sqrt{3} + 2\sqrt{3}$

$5\sqrt{3} + 2\sqrt{3}$

$= (5+2)\sqrt{3}$

$= 7\sqrt{3}$

Try it yourself

Simplify $8\sqrt{7} + 3\sqrt{7}$

STEMTHINKING

Rule 3 $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$

Simplify $\sqrt{\frac{9}{16}}$

$\sqrt{\frac{9}{16}} = \frac{\sqrt{9}}{\sqrt{16}}$

$= \frac{3}{4}$

Simplify $\sqrt{\frac{64}{81}}$

Rationalising Denominator

This means removing the surd from the bottom of a fraction

$$\frac{a}{\sqrt{b}} = \frac{a\sqrt{b}}{\sqrt{b}\sqrt{b}} = \frac{a\sqrt{b}}{b}$$

When a fraction is of the form $\frac{a}{\sqrt{b}}$ ⇨ multiply both the numerator and denominator by $\sqrt{b}$

Simplify $\frac{1}{\sqrt{7}}$

$\frac{1}{\sqrt{7}}$

$= \frac{1 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$

$= \frac{\sqrt{7}}{7}$

Try it yourself

Simplify $\frac{2}{\sqrt{3}}$

STEMTHINKING

Perfect Squares

Determining Maximum and Minimum Values of a Quadratic Function

Recall that:

1. Quadratic equations can be expressed using three forms (which are generally interchangeable):

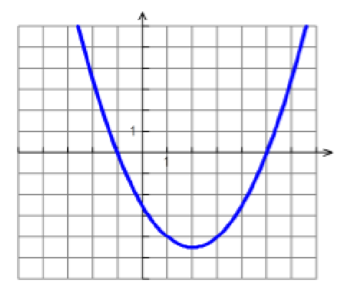
   Standard Form ⇒ $f(x) = ax^2 + bx + c$

   Factored Form ⇒ $f(x) = a(x - s)(x - t)$; zeroes are ___ and ___.

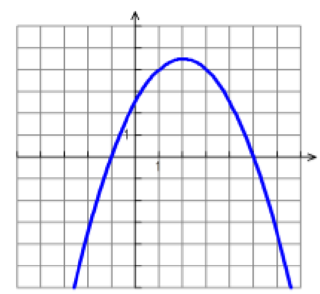
   Vertex Form ⇒ $f(x) = a(x - h)^2 + k$; vertex coordinates are ________.

2. The graph of a quadratic equation is a parabola.

3. The maximum or minimum value of a parabola corresponds to the value of the ___________________ of the vertex.



In $y = a(x - h)^2 + k$, when $a > 0$, the parabola faces up and has a **minimum**.



In $y = a(x - h)^2 + k$, when $a < 0$, the parabola faces down and has a **maximum**.

Completing the Square

Also, recall that:

1. *Completing the Square* is a method used to find the ___________ of a parabola given its standard form equation. This method involves changing the standard form equation into ___________ form.

2. The steps involved are as follows:
   - Group the terms containing the variable, x;
   - Factor out any _________________ of the x^2 term;
   - Inside the brackets, create a perfect square by both adding and subtracting ______;
   - Factor the perfect square;
   - Simplify.

Ex. 1: Find the maximum or minimum value of the function and the value of x when it occurs.

a) $f(x) = -0.4x^2 - 2.4x + 0.6$

b) $f(x) = \frac{1}{3}x^2 - 3x - 4$

Ex. 2: Revenue

A theatre company has 300 season tickets that normally sell for $400 per ticket. It has been determined that for every $20 increase in price, 10 subscribers would not renew their tickets. What price would maximize the revenue from the tickets?

Ex. 3: Maximizing Area

A rectangular field is to be enclosed by a fence and divided into two equal rectangular fields by a fence parallel to one side of the field. If 1200 m of fencing is available, find the dimensions of the field that maximizes its area.

Ex. 4: Unknown Numbers
Two numbers differ by 4. Find the minimum of the sum of their squares, and the two corresponding numbers.

Practice Questions

1. Find the maximum or minimum value of the function and the value of x when it occurs.

   a) $f(x)=x^2+12x-7$ b) $g(x)=-x^2+6x+1$ c) $h(x)=13+x^2-20x$

   d) $f(x)=-4x^2+4x-9$ e) $g(x)=-\frac{1}{3}x^2+2x+4$ f) $h(x)=-2x^2+3x-2$

   g) $f(x)=3x^2-12x+11$ h) $g(x)=-2x^2-4x+1$ i) $h(x)=-36x+6x^2-5$

   j) $f(x)=0.5x^2+x+2$ k) $g(x)=-3x^2+4x$ l) $h(x)=0.5x^2-0.6x$

2. A sportswear store sells baseball caps with the local baseball team's logo on them. Last year, the store sold 600 caps at $15 each. The store manager is planning to increase the price. A consumer survey show that for every $1 increase, there will be a drop of 30 sales a year.
   a) What should the selling price be to maximize the annual revenue?
   b) What is the maximum annual revenue from the caps?

3. Two numbers have a sum of 13.
   a) Find the minimum of the sum of their squares.
   b) What are the two numbers?

4. Determine the maximum area of a triangle, in square centimeters, if the sum of its base and its height is 13 cm.

5. A rectangle has dimensions 3x and 5 – 2x.
   a) What is the maximum area of the rectangle?
   b) What value of x gives the maximum area?

6. A triangle has a height of 2x and a base of 7 – 4x.
   a) What is the maximum area of the triangle?
   b) What value of x gives the maximum area?

7. A rectangular corral is to be built using 70 m of fencing.
   a) If the fencing has to enclose all four sides of the corral, what is the maximum possible area of the corral, in square metres?
   b) In part a), what are the dimensions of the corral with the maximum possible area?
   c) If the corral is built with a wall of a large barn as one side and with the fencing enclosing the other three sides, what is the maximum possible area of the corral, in square metres?
   d) In part c), what are the dimensions of the corral with the maximum possible area?

Answers
1. a) minimum of -43 at x = -6 b) maximum of 10 at x = 3 c) minimum of -87 at x = 10 d) maximum of -8 at x = ½ e) maximum of 7 at x = 3 f) maximum of -7/8 at x = ¾ g) minimum of -1 at x = 2 h) maximum of 3 at x = -1 i) minimum of -59 at x = 3 j) minimum of 1.5 at x = -1 k) maximum of 4/3 at x = 2/3 l) minimum of -9/50 at x = 3/5
2. a) \$17.50 b) \$9187.50 3. a) 84.5 b) 6.5, 6.5 4. 21.125 cm^2 5. a) 9.372 square units b) 1.25
6. a) 3.0625 square units b) 0.875 7. a) 306.25 m^2 b) 17.5 m by 17.5 m c) 612.5 m^2 d) 17.5 m by 35 m

Solving Quadratic Equations

Recall: Quadratic equations can be solved by:

- Solve Directly
- Factoring
- Using the Quadratic Formula (short cut of completing the square)

Ex. 1: Find the roots of each equation by solving directly.

a) $(x + 3)^2 - 25 = 0$

b) $(t - 4)^2 = 56$

Ex. 2: Find its roots by factoring.

a) $x^2 - 10x + 25 = 0$

b) $x^2 - 5x - 24 = 0$

c) $6x^2 - 17x = 14$

Ex. 3: Find its roots using the Quadratic Formula.

a) $x^2 + 3x + 1 = 0$

b) $3x^2 - 2x - 9 = 0$

Solving Quadratic Equations - Applications

Ex. 1: A landscape artist wants to design a rectangular flowerbed measuring 18 m by 12 m. Within the bed there will be a uniform-width border of annuals around the rose bed. The artist wants the rose bed to be exactly half the total area. What is the width of the border? Round to one decimal.

Ex. 2: The height of a triangle is 5 cm less than its base. The area of the triangle is 14 cm^2. Find the dimensions, to the nearest tenth of a centimeter.

Ex. 3: Adding a number to $\frac{1}{3}$ its square gives $\frac{47}{4}$. Express the possible values of the number in simplest radical form.

Solving Quadratic Equations Practice

1. Solve.

   a) $(x + 3)^2 = 9$ b) $(x - 10)^2 - 1 = 0$ c) $(s - 1)^2 = 4$ d) $(y - 4)^2 - 25 = 0$

   e) $4 = \left(x + \frac{1}{2}\right)^2$ f) $\left(x - \frac{1}{3}\right)^2 = \frac{1}{9}$ g) $\left(a + \frac{3}{4}\right)^2 = \frac{9}{16}$ h) $(n - 0.5)^2 = 1.21$

   i) $(x + 0.4)^2 - 0.01 = 0$

2. Solve.

   a) $(x - 4)(x + 7) = 0$ b) $(2y + 3)(y - 1) = 0$ c) $(3z + 1)(4z + 3) = 0$ d) $(2x - 5)(2x + 5) = 0$

3. Solve by factoring. Check solutions.

   a) $x^2 + 3x - 40 = 0$ b) $x^2 - x = 12$ c) $y^2 = 12y - 36$ d) $z^2 - 30 = -z$

   e) $a^2 - 4 = 3a$ f) $b^2 - 5b = 0$ g) $m^2 = 5m + 14$ h) $t^2 = 16$

   i) $4x^2 - 3 = 11x$ j) $4y^2 - 17y = -4$ k) $9z^2 = -24z - 16$ l) $3x^2 = 4x + 15$

   m) $4x^2 = 25$ n) $2m^2 + 9m = 5$ o) $8t^2 = 1 - 2t$ p) $y - 2 = -6y^2$

   q) $6x^2 + 7x + 2 = 0$ r) $5z^2 + 44z = 60$ s) $9r^2 - 16 = 0$ t) $2x^2 = -18 - 12x$

4. Solve using the quadratic formula. Express solutions in fraction form/simplest radical form when needed.

   a) $4x^2 - 12x + 5 = 0$ b) $3y^2 + 5y = 28$ c) $2m^2 = 15 + m$ d) $2z^2 + 3 = 5z$

   e) $3r^2 - 20 = 7r$ f) $4x^2 = 11x + 3$ g) $5a^2 - a = 4$ h) $15 + w - 6w^2 = 0$

   i) $3x^2 + 6x + 1 = 0$ j) $2t^2 + 6t = -3$ k) $4y^2 + 7 - 12y = 0$ l) $m^2 + 4 = -6m$

   m) $0.1 - 0.3m = 0.2m^2$ n) $\frac{x}{2} + 1 = \frac{7x^2}{2}$ o) $y^2 + \frac{y}{6} - \frac{1}{2} = 0$ p) $\frac{t^2}{5} - \frac{1}{2} = \frac{t}{5}$

5. A rectangular lawn measures 7 m by 5 m. A uniform border of flowers is to be planted along two adjacent sides of the lawn. If the flowers that have been purchased will cover an area of 6.25 m^2, how wide is the border?

6. The base of a triangle is 2 cm more than the height. The area of the triangle is 5 cm^2. Find the base, to the nearest tenth of a centimetre.

7. The length of a rectangle is 2 m more than the width. The area of the rectangle is 20 m^2. Find the dimensions of the rectangle, to the nearest tenth of a metre.

8. The sum of an integer and its square is 210. Find the integer.

9. A square of side length x + 1 has an area of 6 square units. Find the value of x, to the nearest hundredth.

10. The sum of two numbers is 14, and their product is 37. What are the numbers
a) in simplest radical form? b) to the nearest thousandth?

11. Subtracting a number from half its square gives a result of 13. Express the possible values of the number in simplest radical form.

12. Two positive integers are in the ratio 1:3. If their sum is added to their product, the result is 224. Find the integers.

Answers

1. a) -6, 0 b) 9, 11 c) -1, 3 d) -1, 9 e) $\frac{-5}{2}, \frac{3}{2}$ f) 0, $\frac{2}{3}$ g) $\frac{-3}{2}$, 0 h) 0.6, 1.6 i) -0.5, -0.3
2. a) -7, 4 b) $\frac{-3}{2}$, 1 c) $\frac{-3}{4}, \frac{-1}{3}$ d) $\frac{-5}{2}, \frac{5}{2}$
3. a) -8, 5 b) -3, 4 c) 6 d) -6, 5 e) -1, 4 f) 0, 5 g) -2, 7 h) -4, 4 i) $\frac{-1}{4}$, 3 j) $\frac{1}{4}$, 4 k) $\frac{-4}{3}$ l) $\frac{-5}{3}$, 3 m) $\frac{-5}{2}, \frac{5}{2}$ n) -5, $\frac{1}{2}$ o) $\frac{-1}{2}, \frac{1}{4}$ p) -1, $\frac{2}{3}$ q) $\frac{-2}{3}, \frac{-1}{2}$ r) -10, $\frac{6}{5}$ s) $\frac{-4}{3}, \frac{4}{3}$ t) -3
4. a) $\frac{-1}{4}, \frac{5}{2}$ b) -4, $\frac{7}{3}$ c) $\frac{5}{2}$, 3 d) 1, $\frac{3}{2}$ e) $\frac{-5}{3}$, 4 f) $\frac{-1}{4}$, 3 g) $\frac{-4}{5}$, 1 h) $\frac{-3}{2}, \frac{5}{3}$ i) $\frac{-3+\sqrt{6}}{3}$, $\frac{-3+\sqrt{6}}{3}$ j) $\frac{-3+\sqrt{3}}{2}$, $\frac{-3-\sqrt{3}}{2}$ k) $\frac{3+\sqrt{2}}{2}$, $\frac{3-\sqrt{2}}{2}$ l) $-3+\sqrt{5}$, $-3-\sqrt{5}$ m) $\frac{-3+\sqrt{17}}{4}$, $\frac{-3-\sqrt{17}}{4}$ n) $\frac{1+\sqrt{57}}{14}$, $\frac{1-\sqrt{57}}{14}$ o) $\frac{-1+\sqrt{73}}{12}$, $\frac{-1-\sqrt{73}}{12}$ p) $\frac{1+\sqrt{11}}{2}$, $\frac{1-\sqrt{11}}{2}$
5. 0.5 m
6. 4.3 cm
7. 3.6 m by 5.6 m
8. -15 or 14
9. 1.45 units
10. a) $7+2\sqrt{3}$, $7-2\sqrt{3}$ b) 10.464, 3.536
11. $1+3\sqrt{3}$ or $1-3\sqrt{3}$
12. 8, 24

The Roots of a Quadratic Function

Recall:

To find the number of roots/zeros that a standard-form quadratic equation has, look at the value of the discriminant, $b^2 - 4ac$. If:

- $b^2 - 4ac > 0$, the equation has 2 real and distinct roots;
- $b^2 - 4ac = 0$, the equation has 2 real but equal roots;
- $b^2 - 4ac < 0$, the equation has 2 imaginary roots.

You may also find the number of roots by actually calculating them; set *f(x)* = 0 in the equation (if this has not already been done) and solve for x by completing the square, factoring, or using the Quadratic Formula.

Ex. 1: Find the number of roots for each of the following equations.

a) $f(x) = 2x^2 - 3x - 5$ b) $f(x) = -5x^2 + x - 2$ c) $f(x) = x^2 - 10x + 25$

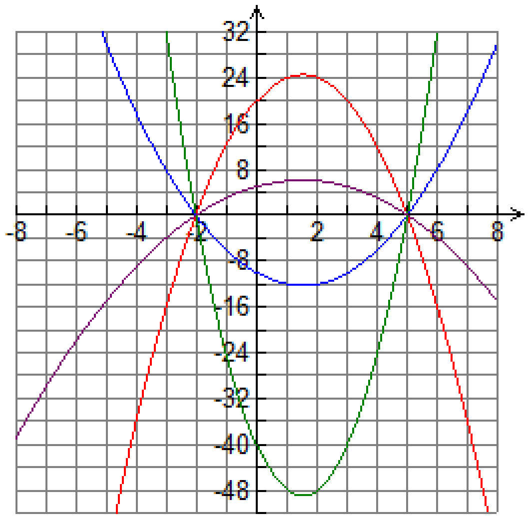
Ex. 2: For $g(x) = 3x^2 - 2x + k$, what are the possible values of *k* if the equation has one root? two roots? no roots?

Ex. 3: For $f(x) = x^2 + kx + 4k - 12$, what are the possible values of k if the equation has one zero? two roots? no roots?

Ex. 4: A market researcher predicted that the profit equation for the first year of a new business would be: $P(x) = -0.3x^2 + 3x - 15$, where x is based on the number of items produced. Will it be possible for the business to break even in its first year?

Families of Quadratic Functions

Family of Parabolas



Common Characteristic

Distinguishing Characteristic

Ex. 1: Find the equation, in **factored form**, for a family of quadratic functions with the given x-intercepts.

a) 4 and 2 b) 0 and -5 c) -3 and 3 d) 6 is the only x-intercept

Ex. 2: Determine the equation, in **standard form**, of the quadratic function with zeros 2 and -3 that passes through (0, 3).

Ex. 3: Determine the equation of the parabola, in **standard form**, with roots $3+\sqrt{5}$ and $3-\sqrt{5}$, and passing through the point (2, -12).

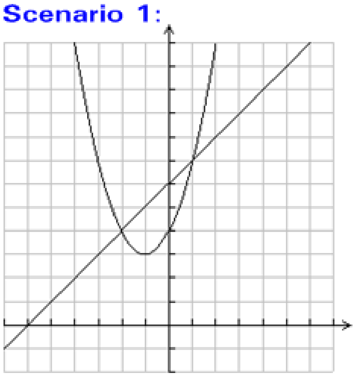
Systems of Linear-Quadratic Equations

Recall:

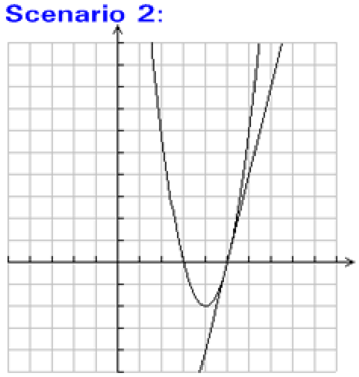
- The graph of a linear equation is a ________.
- The graph of a quadratic equation is a ______________.

The diagrams below illustrate all the possible scenarios, in terms of intersection points, between a line and a parabola.

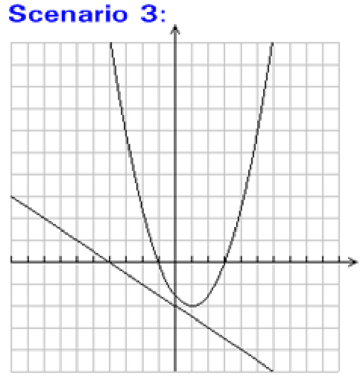
Scenario 1:



Scenario 2:



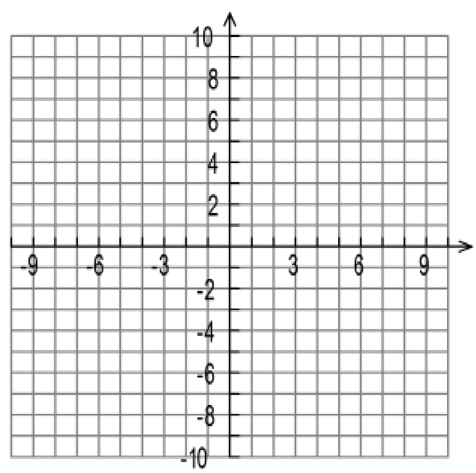
Scenario 3:



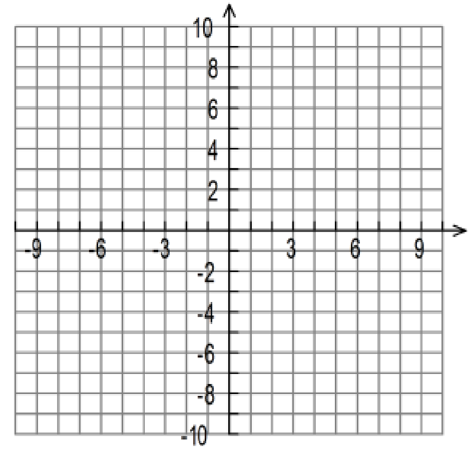
As in the case of a system of two linear equations, the intersection point(s) of a linear equation with a quadratic equation can be found graphically and/or algebraically.

Ex. 1: Find the point(s) of intersection of the given parabola and line. Solve graphically and algebraically.

a) $f(x) = -x^2 + 4x + 2$ and $g(x) = x + 2$



b) $f(x) = x^2 + 2x - 3$ and $g(x) = 4x - 4$



Ex. 2: Determine the number of points of intersection of $f(x) = 3x^2 + 12x + 14$ and $g(x) = 2x - 8$ without solving.

Ex. 3: The revenue equation for a company is $R(t) = -40t^2 + 300t$, where t is the ticket price in dollars. The cost equation is $C(t) = 1600 - 220t$. Determine the ticket price that will allow the company to break even.

Ex. 4: Determine the value(s) of k such that the linear equation $f(x) = -5x + k$ does not intersect the parabola $g(x) = -2x^2 + 3x + 1$.

Unit 3 – Quadratic Functions: Problem-Solving Practice

1. A rectangular prism has dimensions $\sqrt{10}-2$, $\sqrt{10}+2$ and $\sqrt{7}+1$ units. Calculate its volume.
   [$6+6\sqrt{7}$ units^3]

2. A triangle has an area of $\sqrt{3}+4$ units^2, and a base of $\sqrt{5}-1$ units. What is its height?
   [$\frac{4+\sqrt{15}+\sqrt{3}+4\sqrt{5}}{2}$ units]

3. Two numbers have a difference of 8. The sum of their squares is 104. Find the numbers.
   [2 and 10, or -2 and -10]

4. Two numbers have a sum of 11. Find the maximum or minimum product, and the numbers.
   [Max product is $\frac{121}{4}$, and numbers are both $\frac{11}{2}$]

5. The TTC is considering raising the price per fare from the current price of $3.00. Their analysts have determined that, for each $0.25-increase, approximately 50 people less will purchase a fare.

a) If their current ridership is 50000, what is the maximum revenue?
   [$3200450]

b) What should the price per fare be to yield a revenue of $223650?
   [$4.50 or $248.50 - !!!]

6. A flower garden measuring 7 m by 6 m is surrounded by a grass strip of uniform width. If the total area of the garden and the grass strip is 92 m^2, what is the width of the strip? Express your answer(s) in simplified radical form, if necessary.
   [$\frac{-13+3\sqrt{41}}{4}$ m]

7. Determine the value(s) of k so that the quadratic function $f(x)=x^2-kx+3$ has two zeroes. Express your answer(s) in simplified radical form, if necessary.
   [$k<-2\sqrt{3}$ and $k>2\sqrt{3}$]

Review Practice Questions

1. Determine the number of points of intersection for the line $y = x - 4$ and the curve $y = x^2 + 5x + 1$.

2. A water balloon is catapulted into the air from the top of a building. The height, $h(t)$, in metres, of the balloon after t seconds is $h(t) = -5t^2 + 30t + 10$.
   a) What is the maximum height of the balloon?
   b) When does the balloon reach this maximum height?
   c) When does the balloon reach the ground?

3. Barry owns a business that sells parts for electronic game systems. The profit function for his business can be modeled by the equation $P(x) = -0.5x^2 + 8x - 24$, where x is the quantity sold, in thousands, and $P(x)$ is the profit in thousands of dollars.
   a) How many systems does Barry need to sell to maximize profit?
   b) What is his maximum profit?
   c) Determine the number of parts Barry must sell to break even.

4. For what value of b will the line $y = -2x + \text{b}$ be tangent to the parabola $y = 3x^2 + 4x - 1$?

5. For what value(s) of k will the quadratic equation $x^2 - 4x + k = 0$ have two real roots?

6. You want to start up a new business selling video games. With research you determine that if you charge $40, you will sell 500 video games in a week. You also determine that with every $5.00 increase in price you will sell 50 fewer video games in a week. If you can purchase the video games for $20 each, determine the price you should charge to maximize profit and the maximum weekly profit.

7. A math textbook is thrown off of a cliff. The height, $H(t)$, of the book, in meters, can be modeled by $H(t) = -3t^2 + 12t + 20$, where t is the time in seconds the book is in the air.
   a) What is the book's initial height?
   b) What is the maximum height of the book and when does it occur?
   c) How long will the textbook be above 29 m?
   d) When does the book hit the ground (to nearest hundredth of a second)?

8. A rectangular swimming pool measures 6 m by 10 m. The pool is to be surrounded by a deck of uniform width. The total area of the deck is to be the same as the area of the pool. Find the width of the deck accurate to nearest hundredth of a metre.

9. Solve by Quadratic Formula. Express answer as a **simplified radical.**
   a) $-3x^2 - 2x + 2 = 0$ b) $2b^2 + 4b - 3 = 0$

Answers:
1. There are no points of intersection. **2**. a) 55m b) 3sec c) 6.32 sec **3**. a) 8000 units b) $8000 c) 4000 units or 12000 units **4**. b= -4 **5**. k <4 **6**. You should charge $55 to maximize the weekly profit of $12 250. **7**. a) 20m b) the maximum height of the book is 32 m and this occurs after 2 econds. c) The textbook will be above 29 m for 2 seconds. d) The book hits the ground after approximately 5.27 seconds. **8**. The width of the deck is 1.57 m. 9. a) $x = \frac{-1 \pm \sqrt{7}}{3}$ b) $x = \frac{-2 \pm \sqrt{10}}{2}$